Removing Factorial Authenticity

Binomial prime-power certificates for Diophantine consistency

AI-assisted mathematical research

Prepared for Vladimir Reshetnikov and the ProveIt project

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Abstract

We construct a fixed, fully emitted quartic polynomial over the unrestricted integers whose unsolvability is provably equivalent in ZFC to Con(ZFC), using the repository's numerical proof calculus C_{222}. The principal endpoint has 242 variables, 178 quadratic residuals, 5,018 collected monomials, and coefficient height 7,968. Every residual is printed in this report; the accompanying JSON files also supply the complete collected polynomial.

The principal mathematical change is to remove the assertion that a certificate variable equals $t!$. For a sufficiently large ordinary integer R, the binomial coefficient $B = \binom{R+t}{t}$ itself retains a prime-power selector for every one of t rows. Its factors need not be pairwise coprime. Conversely, every prime of B can be assigned a compatible row, which closes the completeness direction. A separate large-evaluation argument recovers proof-label multiplicities without coprime numerical label weights. Thus an explicit power replaces both uses of the old factorial. An odd-square radix, a cofinal square choice for the beta base, and exact quadratic projections complete the smaller equation.

The new arithmetic arguments are ordinary unbounded proofs. Finite programs check the literal source edits, elimination identities, coordinate changes, residual transformations, and every collected coefficient. The logical C_{222} calibration and unchanged Pell compiler are identified inherited results. This release is not an end-to-end proof-assistant formalization, a proof of consistency, or a claim of a global simplicity optimum.

Comparison snapshot: `3ecd6a6cf53d47519228328567f45fb7ee9781d9`.
All final unknowns in the principal equation range over $\mathbb{Z}$.

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1 The result and the meaning of "simpler"

The primary mathematical object is

$$P(x_0,\dots,x_{241}) = \sum_{j=0}^{177} R_j(x_0,\dots,x_{241})^2 = 0, \qquad \boldsymbol{x} \in \mathbb{Z}^{242}. \tag{1}$$

Appendix A gives every coefficient of every quadratic R_j. The delivered `direct/principal/equation.json` is the machine-readable definition; the neighboring `expanded_polynomial.json` is its completely collected expansion. An intermediate register in the construction is a genuine variable only when it remains in this final coordinate list.

Theorem 1.1 (Consistency calibration)**.** *For the fixed polynomial* (1)*, the new arithmetic proofs in this report, composed with the pinned C_{222}/ZF/ZFC calibration, establish*

$$\mathrm{ZFC} \vdash \left(\neg\exists \boldsymbol{x} \in \mathbb{Z}^{242}\ P(\boldsymbol{x}) = 0\right) \iff \mathrm{Con(ZFC)}.$$

No consistency assumption is used to prove this equivalence.

The direct route compiles numerical proof rules. It does not simulate K215, specialize Jones's universal polynomial, or contain a large hidden machine coefficient. The coefficients are the ordinary small integers printed in the residual appendix. This distinction matters for inspection and formalization, even though existing native constructions use many fewer variables.

1.1 Exact comparison

Equation over $\mathbb{Z}$	Variables	Rows	Degree	Terms	Height
Matched parent	252	182	4	5,340	128,446
Principal	242	178	4	5,018	7,968
Dense companion	241	177	4	12,774	7,968

Here *terms* means nonzero monomials after complete collection, and height is the largest absolute coefficient in that expansion. The principal equation saves ten variables and four residuals relative to its matched parent. Its support decreases by 322 monomials and its coefficient height by approximately 93.80%. The maximum absolute coefficient in a quadratic residual is 64 for both new endpoints.

The baseline is the latest common-remainder direct quartic in the eight incoming submissions present at the comparison commit [2, 1]. The separate centered-residue and residual-basis submissions supply useful independent transformations [3, 4]. We credit these transformations rather than presenting their composition as a new theorem. The factorial-free source is a change to the arithmetic architecture, and is the principal contribution of this report.

There is no metric-independent smallest equation. The inspected repository already has nine-positive-variable equations, ten-integer equations using a different reduction, and a fifteen-positive-variable 66-operation circuit. Those are different tradeoffs in domain, degree, coefficient description, and evaluation cost. Our improvement concerns the direct small-coefficient quartic family. The lower-dimensional dense companion deliberately spends more support to remove one coordinate. Neither a global minimum nor literature-wide priority is asserted.

2 Why the old factorial can be removed

In the inherited direct architecture one integer $F = t!$ serves two jobs. First, the weights $1 + jF$, for labels $1 \le j \le t$, are pairwise coprime, so their prime valuations recover proof-label multiplicities. Second, $\binom{F^2+t}{t}$ decomposes into pairwise coprime reciprocal factors $1 + F^2/i$, providing a CRT slot for each proof row. Certifying $F = t!$ requires a self-referential ceiling identity and its supporting bounds.

We replace the two jobs independently. The label product contains only boundedly many factors, so equality at a sufficiently large integer can be lifted to an identity of polynomials. For row selection, pairwise coprime reciprocal factors are stronger than needed: appropriate prime powers inside one binomial coefficient already provide all row selectors. The remaining issue is completeness at *every* prime of that coefficient; Lemma 3.2 handles it explicitly.

The centered-residue predecessor already considered coefficient separation for the label product, but identified the independent factorial dependence of the CRT carrier as an obstacle [3]. The prime-power compiler below removes that remaining dependence. This is the substantive new architectural step.

The new arithmetic choice is

$$d = 2(t+1), \qquad F = d^t = 2^t(t+1)^t. \tag{2}$$

The product argument below first recovers equal factor counts, so the smaller bound $F > (t+1)^t$ suffices. This is a semantic description of an ordinary polynomial certificate. The old power request at index t is reused, so the number of power requests remains eighteen. Variable exponentiation is not an operation in the final polynomial.

3 Prime powers in a binomial modulus

Set $\Lambda_t = \operatorname{lcm}(1, \dots, t)$ for $t \ge 1$. Write $v_p(n)$ for the exponent of the prime p in a positive integer n.

Lemma 3.1 (A classical denominator bound)**.** *For $1 \le i \le t$,*

$$t\binom{t-1}{i-1} \mid \Lambda_t.$$

Proof. For $d = p^a$, the summand in the factorial valuation of the left side is

$$\left\lfloor \frac{t}{d} \right\rfloor - \left\lfloor \frac{i-1}{d} \right\rfloor - \left\lfloor \frac{t-i}{d} \right\rfloor.$$

It is zero or one, because $(i-1) + (t-i) = t-1$, and it vanishes for $d > t$. The valuation is therefore at most $\lfloor \log_p t \rfloor$, exactly the valuation of Λ_t. □

This divisibility is classical and follows also from the lcm identity for binomial rows discussed by Farhi [6]. Its use below is credited accordingly; it is not claimed as a new number-theoretic identity.

Lemma 3.2 (Every prime has a compatible row)**.** *Let $R, t \ge 1$, $B = \binom{R+t}{t}$, and $p^e \| B$. There is an $i \in \{1, \dots, t\}$ for which $p^e \mid R + i$.*

Proof. Let $v = \max_{1 \le i \le t} v_p(R+i)$. Legendre's formula gives

$$e = \sum_{a \ge 1} \left(\left\lfloor \frac{R+t}{p^a} \right\rfloor - \left\lfloor \frac{R}{p^a} \right\rfloor - \left\lfloor \frac{t}{p^a} \right\rfloor \right).$$

Each summand is zero or one. Above $a = v$ the interval contains no multiple of p^a; necessarily $p^a > t$, since any t consecutive integers contain such a multiple if $p^a \le t$. All those summands are zero, so $e \le v$. A row attaining the maximum works. □

Lemma 3.3 (Every row has a sufficiently large selector)**.** *Let $R \ge \Lambda_t^2$ and $B = \binom{R+t}{t}$. For each $i \in \{1, \dots, t\}$ there is a prime power $p^e \| B$ such that*

$$p^e \mid R + i, \qquad p^e > t.$$

Proof. Because $R+i > \Lambda_t^2$, the integer $R+i$ does not divide Λ_t^2. Hence some prime p has $v = v_p(R+i) > 2a$, where $a = v_p(\Lambda_t) = \lfloor \log_p t \rfloor$. In particular $p^v > t$. For $j \ne i$, the valuation of $R + j$ is therefore the valuation of $j - i$. Put

$$d = v_p\left(t\binom{t-1}{i-1}\right).$$

The valuation of B is exactly

$$e = v + v_p((i-1)!(t-i)!) - v_p(t!) = v - d.$$

Lemma 3.1 gives $0 \le d \le a$. Consequently $e \ge v - a > a$, so $p^e > t$, while $e \le v$ gives $p^e \mid R + i$. This also covers $p > t$, for which $a = d = 0$. □

The two lemmas have different purposes. A row selector proves soundness for that row. Lemma 3.2 supplies a compatible assignment at every prime of B in the converse construction, including primes that were not chosen as row selectors. Omitting the latter step would leave a real completeness gap.

3.1 A small example

Take $t = 3$, $R = 37$. Here $\Lambda_3^2 = 36$ and

$$B = \binom{40}{3} = 9880 = 2^3 \cdot 5 \cdot 13 \cdot 19.$$

The consecutive factors are $38, 39, 40$, which are not pairwise coprime. Their full prime powers in B can be assigned as follows:

Prime power in B	Compatible factor	Row index i
19	38	1
13	39	2
8 and 5	40	3

Every row has a selector greater than 3, and all primes of B have a compatible row. No factorization into $1 + F^2/i$ is used.

4 A bounded-row certificate without factorial authenticity

We first recall the exact one-sided range fact used in the repository.

Lemma 4.1 (Binomial range)**.** *If p is prime, $A \in \mathbb{N}$, $k, e \ge 1$, and $p^e \mid \binom{A}{k}$, then $A \bmod p^e < k$. The convention is $\binom{A}{k} = 0$ for $A < k$.*

Proof. The case $A < k$ is immediate. Otherwise, if none of the k consecutive integers $A-k+1, \dots, A$ is divisible by p^e, none is divisible by a higher power either. The valuation formula for the binomial is a sum of zero-one terms, with all terms from level e onward zero. The total is at most $e-1$, a contradiction. Thus $A-r$ is divisible by p^e for some $0 \le r < k$; its ordinary residue is at most r and hence is less than k. □

Theorem 4.2 (Binomial CRT compiler)**.** *Let $t \ge 1$, $R \ge \Lambda_t^2$, $B = \binom{R+t}{t}$, and $N = B^K$. Let $b_1, \dots, b_d \in \mathbb{N}$, with $K \ge \max(t, b_1+1, \dots, b_d+1, 1)$. Suppose a fixed integer polynomial $\mathcal{L}$ satisfies*

$$0 \le \mathcal{L}(i, \boldsymbol{w}) < 2^K \quad (0 \le i < t,\ 0 \le w_j \le b_j).$$

Then the following are equivalent:

1. *Every row $i < t$ has a bounded tuple $\boldsymbol{w}$ with $\mathcal{L}(i, \boldsymbol{w}) = 0$.*
2. *There are nonnegative $C, A_1, \dots, A_d$ satisfying*

$$B \mid C + R + 1, \tag{3}$$

$$N \mid \binom{C}{t}, \tag{4}$$

$$N \mid \binom{A_j}{b_j+1} \quad (1 \le j \le d), \tag{5}$$

$$N \mid \mathcal{L}(C, \boldsymbol{A}). \tag{6}$$

In the forward construction the representatives may all be chosen in $[1, N^2]$. Soundness does not require that upper bound.

Proof. For soundness fix $i \in \{1, \dots, t\}$ and choose p^e from Lemma 3.3. By Lemma 4.1, the index range gives $C \equiv s \pmod{p^{Ke}}$ with $0 \le s < t$. Reduction of (3) modulo p^e gives $s \equiv i-1 \pmod{p^e}$. Since $p^e > t$, we have $s = i-1$ exactly. Each other range condition supplies $A_j \equiv w_j \pmod{p^{Ke}}$ with $0 \le w_j \le b_j$. Therefore p^{Ke} divides $\mathcal{L}(i-1, \boldsymbol{w})$. The latter lies in $[0, 2^K)$ and $p^{Ke} \ge 2^K$, so it is zero.

Conversely choose a valid tuple for each row. For *every* prime $p \mid B$, write $e = v_p(B)$ and choose a compatible i by Lemma 3.2. Prescribe, modulo p^{2Ke},

$$C \equiv i-1, \qquad A_j \equiv w_j^{(i-1)}.$$

These moduli are pairwise coprime and have product N^2. CRT supplies positive representatives at most N^2. The raw index and local congruences follow. For a range lower index $k \le K$, its selected residue w is in $[0, k-1]$. If $A < k$ its binomial is zero; otherwise the falling factorial for $\binom{A}{k}$ has the factor $A-w$ divisible by p^{2Ke}. Division by $k!$ loses at most $v_p(k!) \le k-1 < K \le Ke$. Thus the binomial is divisible by p^{Ke}. This holds for every prime of B and for every range. □

Why a size hypothesis cannot simply be dropped. Let $t = 2$, $R = 1$, $B = 3$, $K = 2$, $N = 9$, $C = 1$, and $\mathcal{L}(i) = (i-1)^2$. Then all three applicable global congruences hold: $3 \mid C+2$, $9 \mid \binom{C}{2} = 0$, and $9 \mid \mathcal{L}(C) = 0$. But row 0 is false. The missing row's prime factor was cancelled by the binomial denominator. This explicit counterexample is included in the finite regression script.

5 Proof-label balance from a polynomial identity

Let a calibrated Horn rule have one conclusion and at most two premises. Pad unused premise slots by the freely derivable truth code 0. Suppose there are t rows and all their labels lie in $[0, t]$. For a fixed

positive integer F put

$$W_j = 1 + jF, \qquad W_0 = 1.$$

Lemma 5.1 (Large evaluation recovers multiplicities)**.** *If $F > (t+1)^t$, then*

$$\prod_{i<t} W_{v_i} = (1+F) \prod_{i<t} W_{p_i} W_{q_i} \tag{7}$$

implies, at each positive label j, that its conclusion multiplicity minus its premise multiplicity is 1 for $j = 1$ and 0 otherwise.

Proof. First discard all unit factors with label zero. The left side has at most t nonunit factors; the right side has at most $2t+1$. If these counts differ, their minimum r is at most t. The side with more factors is at least $(F+1)^{r+1}$, whereas the side with fewer is at most $(1+tF)^r \le t^r(F+1)^r$. Since $F + 1 > t^t \ge t^r$, this is impossible. Thus the factor counts agree and both are at most t.

Replace F by an indeterminate z. Each side now has nonnegative coefficients and coefficient sum at most $(t+1)^t < F$. Thus every coefficient of the difference has absolute value less than F.

A nonzero integer polynomial with that coefficient bound cannot vanish at F: take its lowest nonzero coefficient a_r, divide its evaluation by F^r, and reduce modulo F. This would force $F \mid a_r$ with $0 < |a_r| < F$. Therefore (7) is an identity in $\mathbb{Z}[z]$. Distinct positive labels give distinct irreducible linear factors $1 + jz$ over $\mathbb{Q}$, with roots $-1/j$. Unique factorization gives the required multiplicities. Label 0 contributes only the unit. □

Pairwise coprimality of the *evaluated* integers W_j is not required. The choice (2) has the needed strict bound.

Lemma 5.2 (Single-conclusion flow)**.** *If the balance in Lemma 5.1 is supplied by valid single-conclusion Horn rules and truth is derivable, then label 1 is derivable.*

Proof. Take the finite closure of truth under the finitely many rule instances in the certificate. If 1 is outside this closure, every rule whose conclusion is outside has at least one premise outside. Summing conclusions minus premises over outside labels is therefore at most zero. The balance makes that sum one. This contradiction gives a finite derivation. Repeated premises and cyclic rule multisets cause no problem. □

This last lemma is the inherited commutative-certificate argument, included to make clear why an unordered balance does not permit circular false proofs [5]. The single-conclusion restriction is essential.

6 The selected arithmetic front

All expressions in this section describe values of explicitly certified polynomial registers. The final system itself is entirely quadratic. Let $t = 16 + h^2$ and retain the same C_{222} local rule polynomial, range columns, semantic anchor, and eighteen power requests as the parent. The selected changes are

$$d = 2(t+1), \qquad F = d^t, \tag{8}$$

$$s = (2v+1)^2, \qquad y = Fs, \tag{9}$$

$$q = (2z+1)^2, \qquad H = 2 + C + \sum_{j=1}^{8} A_j. \tag{10}$$

Here C and the eight A_j are the actual CRT representatives; some stored source coordinates are these values minus one. The distinction is retained explicitly in the mutation ledger. The former factorial ceiling rows and $y = V_0(V_0 - 1)$ are deleted. The variable x is unrestricted; there is *no* parity equation for x in the selected source.

The unchanged endpoint and arithmetic requests include

$$x = F + 1 + (y + 1)u_0, \tag{11}$$
$$x = 1 + [1 + (t + 1)y]u_t, \tag{12}$$
$$Bh_B = C + F^2 + 1, \tag{13}$$
$$N = B^x, \qquad q = (BH)^x. \tag{14}$$

The digit called the "factorial digit" in old filenames now has only the intended meaning

$$B = \binom{F^2 + t}{t}. \tag{15}$$

It no longer authenticates a factorial. Old variable names retained in data provenance do not change this mathematical interpretation.

6.1 The unchanged Pell interface and signed bases

For an integer center A, let $X_k(A), Y_k(A)$ be the Pell polynomials defined by

$$X_0 = 1, \quad Y_0 = 0, \qquad X_{k+1} = AX_k + (A^2 - 1)Y_k, \quad Y_{k+1} = X_k + AY_k,$$

and $X_{-k} = X_k$, $Y_{-k} = -Y_k$. The retained signed square frame, semantic anchor, and pooled small-residue proof establish these exact coordinates at every requested signed index. They bound every actual base, index, and output atom in absolute value strictly below W, with $W \geq 258$ and $|A| > W^W$. These are properties of the unchanged graph; the new base $2(t+1)$ is explicitly inserted into its atom list. The relevant ordinary proofs are retained in `direct/parent/` and identified in the source ledger [2, 3].

For a nonzero signed base n define $T_n = 2An - n^2 - 1$. The certified power congruence and the recurrence imply

$$m \equiv n^k \pmod{T_n} \quad (k \geq 0), \qquad mn^{-k} \equiv 1 \pmod{T_n} \quad (k < 0).$$

The modulus satisfies $|T_n| > W^W + W$. For $|n| \geq 2$ the corresponding absolute errors are less than the modulus. A negative k would force the impossible integer equality $mn^{-k} = 1$. Thus $k \geq 0$ and $m = n^k$. For $n = \pm 1$ the same recurrence gives exactly $m = (\pm 1)^k$ for signed k. A base of zero is not covered by this argument, and is excluded explicitly below when needed.

7 Soundness without a circular sign argument

The order of recovery is essential. In particular, neither the sign of B nor the meaning of its binomial digit is assumed in an earlier step.

7.1 Recover the power, representatives, and positive exponent

The request (8) has known positive base and $t \geq 16$, so it gives $F = (2(t + 1))^t > 2^{32}$ independently of every digit. Equation (9) gives $y \geq F > 0$. Equation (10) gives $q \geq 1$, so every numerator request at

base $q+1 \geq 2$ forces its actual upper index to be nonnegative. Hence $C, A_j \geq 0$ and $H \geq 2$. The raw index row (13) excludes $B = 0$ because its right side is positive. Therefore $|BH| \geq 2$.

The signed-base request $q = (BH)^x$ now forces $x \geq 0$ and the actual power equality, even if the tentative value of B is negative. The case $x = 0$ is impossible in (12): its coefficient $1 + (t+1)y$ is greater than one and cannot divide -1. Thus $x > 0$. If $u_0 \leq -1$, (11) gives $x \leq F - y \leq 0$, also impossible. Consequently

$$x \geq F + 1 \geq 3, \qquad x \geq 2 + (t+1)y \geq 2 + (t+1)F. \tag{16}$$

The second inequality follows from $u_t \geq 1$ in the last endpoint.

Since $q > 0$ and $q = (BH)^x$, we now know

$$q = |BH|^x \geq 2^x > 1, \qquad |B| \leq |BH| < q/2. \tag{17}$$

All powers at base q are therefore genuine, and $t, t+1, y+1 \leq x$. The exact digit for B uses no range factor N, so the latter's positive interpretation can wait until $B > 0$ has been proved.

7.2 Authenticate the binomial from small residues

For $0 \leq j \leq t$ a convenient elementary bound is

$$\binom{F^2+t}{j} \leq (F^2+t)^t \leq (F^2+F)^t \leq 2^{t(F-1)}.$$

Here $F \geq t$ and $F^2 + F \leq 2^{F-1}$ for $F \geq 7$, by a direct base case and induction. Combining with (16) gives

$$\frac{2^{t(F-1)}}{q} \leq 2^{-F-t-2}, \qquad \frac{2^{t(F-1)}}{q-1} < \frac{1}{32}.$$

Thus the actual low remainder of $(q+1)^{F^2+t}$ modulo q^t is less than $q^t/32$, and the coefficient $C_* = \binom{F^2+t}{t}$ is less than $q/32$.

The retained common-remainder construction uses $M = q^x$. For each digit denominator $D_i \mid M$, it replaces a remainder r_i by $\rho_i = (M/D_i)r_i$ and imposes

$$M^2 = 1 + 4\sum_{i=1}^{11} \rho_i^2 + \alpha^2 + \beta^2 + \gamma(\gamma+1). \tag{18}$$

The last three terms are nonnegative for integers, so $|r_i| < D_i/2$. The scaled digit row makes r_i an integer congruent to its actual remainder. If that actual remainder is in $[0, D_i/32)$, the difference has absolute value below D_i, and therefore is zero. This also explains why the normalizing ratio variables do not introduce rational solutions.

Apply this first to the exact digit for B. It gives $B \equiv C_* \pmod{q}$. By (17), $|B - C_*| < q/2 + q/32 < q$. Therefore

$$B = C_* > 0.$$

Only at this point do we use $B \geq 2$, $N = B^x > 0$, and the usual range interpretation. This signed reconstruction replaces the old output-slack positivity guard.

7.3 Authenticate every remaining range

For an ordinary upper representative $a \geq 0$ and a required index $0 \leq j \leq k \leq x$,

$$\binom{a}{j} \leq \max(1, a)^k \leq H^x.$$

This includes $a = j = 0$, a case that an a^k bound would miss. The derived label has the inherited bound $h_* \leq H^\ell$ with $\ell = 4194304$. The constant shift in H still gives the required bound for zero representatives. The growth in (16) implies $\ell(t+1) < x$, so the derived-label coefficients also lie below H^x. Because $q = B^x H^x$ and $B \geq 2$,

$$\frac{H^x}{q-1} < \frac{2}{2^x} < \frac{1}{32}.$$

Every remaining low remainder is therefore selected correctly by (18). The independent-column divisibilities are those of Theorem 4.2. There is also a range for a polynomially derived label $J(\boldsymbol{A})$, rather than a new independent column. For a selected prime power, that range gives $J(\boldsymbol{A}) \equiv r \pmod{p^{xe}}$ with $0 \leq r \leq t$. The column congruences imply $J(\boldsymbol{w}) \equiv r \pmod{p^{xe}}$. The inherited local ledger bounds this nonnegative derived value strictly below $2^x \leq p^{xe}$; hence $J(\boldsymbol{w}) = r \leq t$. This is the precise extension of the compiler used for the derived label.

The unchanged local majorant is $0 \leq \mathcal{L} < \mathcal{R}^E$, where $\mathcal{R} = 2 + t + x + y + F$ and $E = 8388608 = 2^{23}$. From $y \geq F$ and (16), $\mathcal{R} < 2x$. Also $x > 2^{32}$, and the elementary dyadic estimate $2^{23} \log_2(2x) < x$ gives

$$\mathcal{L} < (2x)^E < 2^x.$$

The exponent $K = x$ therefore satisfies every range and evaluation budget. Since $F > t! \geq \Lambda_t$, the radius $R = F^2$ satisfies the new CRT hypothesis. Every local row is recovered.

7.4 The boundary case $y = F$

The cofinal square parametrization permits $y = F$ algebraically. It must not be silently treated as the old strict bound. After bounded row decoding, the first endpoint has remainder zero modulo $y + 1$, because $F + 1 = y + 1$. Thus the initial accumulator is zero. Each local update is

$$S_{i+1} W_{v_i} = S_i W_{p_i} W_{q_i},$$

with every weight positive, so zero propagates to the final accumulator. The last endpoint has ordinary remainder one and forces that final accumulator to be one, a contradiction. Hence every actual zero has $y > F$. Now $F + 1 \leq y$, and the first endpoint restores the intended initial accumulator $S_0 = F + 1$.

All accumulators are positive by induction. Multiplying their updates and cancelling gives (7). Lemma 5.1 and the single-conclusion flow argument yield a finite C_{222} refutation. The retained logical calibration then yields a finite syntactic ZFC refutation.

8 Constructive completeness

Start with a finite calibrated C_{222} refutation. Expand it to a finite proof tree, and choose a horizon $t = 16 + h^2$ beyond its size, all its schema parameters, and all required local label sums. This choice precedes all large arithmetic witnesses. Define F by (2), and construct the pending-multiset accumulators using weights $1 + jF$. Truth padding gives exactly t rows.

Let O be the odd part of $t!$. Choose an arbitrarily large odd multiple w of O and set $y = Fw^2$. Because

$$v_2(F) = t\big(1 + v_2(t+1)\big) \geq t > v_2(t!),$$

y is an even multiple of $t!$. Choose w large enough that $y > F$ and every accumulator is at most y. The beta moduli $1 + (i+1)y$, $0 \leq i \leq t$, are then pairwise coprime: a common prime would divide $j - i \leq t$ and hence y, while also dividing $1 + (i+1)y$, impossible. They are all odd. CRT supplies a common beta code x; adding their product changes its parity, so choose x even. The genuine endpoint quotients satisfy (16), and the local beta quotients are strictly below x because their moduli are at least two.

Define $R = F^2$, $B = \binom{R+t}{t}$, and $N = B^x$. The new binomial CRT construction supplies all actual representatives modulo N^2. The integer B is odd: for $1 \leq i \leq t$, $v_2(R) > \lfloor \log_2 t \rfloor$ gives $v_2(R+i) = v_2(i)$, so their valuations cancel exactly against $t!$.

Consequently the representative classes modulo N^2 have free parity. Choose the dominant schema representative a odd, and the other eight representatives even. Take a in $[2N^2, 4N^2)$ by adding suitable multiples of the odd modulus. The inherited shell constructors give $h_* = 1 + a + b \pmod 2$ and $h_* \geq 2a^2$, so the semantic anchor is even and dominates the other indices. More explicitly, choose the other eight positive representatives at most $2N^2$. The front indices satisfy $F^2 + t \leq B < N^2$ and $x, y+1, t < N^2$, whereas $h_* \geq 8N^4 > 2\max(\text{other indices})$. The derived-label range is preserved because polynomial evaluation respects the CRT congruences modulo p^{2xe}, and its denominator loses at most $v_p((t+1)!) \leq t < x$ powers of p. Meanwhile $H = 2 + \sum a_j$ is odd. Since B is odd and x is even,

$$q = (BH)^x$$

is an odd square and has the parametrization in (10). All four denominator groups divide q^x. The genuine normalized remainders satisfy $0 \leq \rho_i < q^x/32$, so

$$4\sum_{i=1}^{11} \rho_i^2 < \frac{44}{1024}(q^x)^2 < (q^x)^2.$$

The nonnegative integer gap in (18) is represented by $\alpha^2 + \beta^2 + \gamma(\gamma+1)$, using the standard three-square representation retained in the parent. Explicitly, for any gap $g \geq 0$, Legendre's three-square theorem represents $4g+1$; reduction modulo four makes exactly one of its three squares odd. Writing the even roots as $2\alpha, 2\beta$ and the odd root as $2\gamma + 1$ gives the stated form.

The unchanged constructive Pell frame and anchor supply the finite family of genuine powers. The new request has index t, which was already present; no new index-compatibility premise has appeared. This constructs a zero of the changed arithmetic source. Exact integral projections and reversible row/coordinate operations transport it to (1). Completeness uses factorials in the *proof of existence of a suitable multiple*; no unknown is required to equal a factorial in the certificate.

9 From the arithmetic source to a literal quartic

The new source keeps the calibrated local C_{222} polynomial, its eighteen power slots, and the common-remainder transformation. All new definitions are quadratic register rows. The single variable power in (8) is compiled by the retained Pell graph at the existing index t. Thus every final residual is an ordinary integer polynomial of degree at most two; summing their squares gives degree at most four. The emitted expansion has nonzero degree-four coefficients, so its degree is exactly four.

The compiler performs three kinds of finite transformations. First, it substitutes variables whose defining row is monic and whose substitution keeps every remaining row quadratic. The deleted variable is

restored uniquely by its integer polynomial. Second, it exposes three already valid affine consequences by exact row identities and projects their variables. A fourth such projection gives the denser companion. Third, it applies recorded integral coordinate translations and elementary residual-row operations. A row operation replacing R_i by $R_i + nR_j$ is reversed by subtracting nR_j; it preserves the common zero set even though it changes the sum-of-squares polynomial.

These facts are checked coefficient by coefficient. The row-reduction search is a heuristic used only to find a useful basis; its frozen finite operation list is the certificate. Reproducing that list does not require rerunning the search. No random modular agreement is treated as an identity proof.

10 Logical calibration and verification boundary

The logical target is the repository's fixed numerical calculus C_{222} with truth code 0 and falsity code 1. The retained named interfaces are

`KEndpoint.c222_iff_zf_refutation`,
`ConstructibleChoice.zf_refutation_iff_zfc_refutation`.

The first calibrates the numerical rules with syntactic ZF refutability; the second uses constructible choice for ZF/ZFC equiconsistency. Ordinary proof sources and the relevant interface source files are retained with their provenance. Their complete transitive proof-assistant closure was not rebuilt in this task.

The new proofs above establish, in finite arithmetic formalizable in ZFC, both implications between a zero of the emitted polynomial and a finite C_{222} refutation. Composing the identified calibration gives

$$\mathrm{ZFC} \vdash \big(\exists \boldsymbol{x}\ P(\boldsymbol{x}) = 0\big) \longleftrightarrow \big(\exists n\ \mathrm{Proof}_{\mathrm{ZFC}}(n, \bot)\big).$$

Negation gives Theorem 1.1. This is an ordinary mathematical provability argument, not an emitted object-language proof string. The distinction between an unbounded proof and a finite artifact check is preserved throughout the release.

10.1 Finite audit results

The normal audit first reconstructs the inherited 252-coordinate source and expansion. For the new source it checks 18 Horn factorization edits, 202 unit-monic projections, and the removal of 114 coordinates that do not occur in the retained system. The final maps contain 6 integral coordinate shears and 5 affine-exposure row operations. The principal then uses 3 affine projections and 34 final row operations; the companion uses 4 and 36, respectively. The checker independently collects every coefficient of both quartics. It imports no producer module. Seven deliberate corruptions are rejected. The new finite checks also pass with Python's `-O` flag, so their validity does not depend on enabled `assert` statements.

A separately authored checker reconstructs the eighteen power requests, thirty-seven distinct frame atoms, fourteen principal Pell pairs, five base moduli, and the derived consecutive-index pair in the changed raw source. It too passes normally and with `-O`.

A fresh reconstruction reproduces eighteen JSON files byte for byte, checks twelve endpoint alias files, and verifies six readable exports. The small-case number-theory script separately exercises the new coverage, range, CRT precision, label, parity, and cofinality arguments, including an explicit counterexample without the radius hypothesis. These finite regressions supplement the unbounded proofs; they are not substitutes for those proofs or for a formal proof-assistant development.

11 What is new, inherited, and still open

The new contribution is the explicit replacement of factorial authenticity by a binomial prime-power compiler with two-sided row coverage, together with the compatible proof-label and signed-source changes. The sharpened Λ_t^2 threshold is proved here using a classical lcm divisibility. The polynomial-evaluation device itself is elementary; its use with the bounded C_{222} label product removes the need for coprime evaluated weights. The cofinal odd-square beta base adds a further semantic saving, with the $y = F$ exceptional branch explicitly excluded.

The numerical calculus, signed Pell frame, range lemma, common-remainder normalization, and affine residual exposures have identified predecessors. The archive records their exact source boundaries. This report does not claim that an exhaustive literature search establishes worldwide novelty. It establishes a complete new arithmetic source relative to the inspected repository and an exact smaller direct quartic.

Several questions remain worthwhile. The selector threshold $R \geq \Lambda_t^2$ is sufficient and need not be optimal. A row-specific threshold using $\Lambda_t\, t\binom{t-1}{i-1}$ may reduce it further, although this would not presently lower the emitted graph size. The largest remaining arithmetic cost is the shared Pell certification, so a new finite equation for its simultaneous powers would have greater impact than another local row-basis search. Finally, an end-to-end formalization should start with the prime-power coverage theorem and signed bootstrap: they are compact statements with explicit exceptional cases and clear interfaces to the existing graph.

12 Files, reproduction, and integration

The archive is intended as a new contribution under `Computability/DiophantineConsistency/con-zfc-diophantine-forms`, or initially in `docs/incoming`. It includes this source and its compiled PDF, the literal principal and companion equations, all collected coefficients, construction and checking programs, frozen transformation certificates, mathematical reviews, retained ordinary proof dependencies, licenses, and a checksum manifest.

Run `python3 -B verify.py` from the archive root for the selected finite checks. They use Python's standard library and do not need a network or a consistency oracle. The evaluator accepts a supplied integer tuple; it does not attempt a root search. Rebuild this report with two runs of `pdflatex -interaction=nonstopmode -halt-on-error main.tex` from the `report/` directory. Different TeX versions can change PDF bytes without changing the polynomial or mathematics.

References

[1] V. Reshetnikov and ProveIt contributors, *ProveIt: DiophantineConsistency*, comparison commit `3ecd6a6cf53d47519228328567f45fb7ee9781d9`, 25 October 2026. https://github.com/VladimirReshetnikov/ProveIt/tree/3ecd6a6cf53d47519228328567f45fb7ee9781d9/Computability/DiophantineConsistency.

[2] ProveIt incoming research, *Common remainder certificates for Diophantine consistency*, `ConZFC_Common_Remainders_Radix47_2026-10-25.zip`, `direct/proofs/NORMALIZED_REMAINDERS_PROOF.md` and its retained orientation/strict-budget proof sources. Available in `docs/incoming` at the comparison commit; exact hashes are in this archive's source ledger.

[3] ProveIt incoming research, *Centered residue certificates and reusable Pell denominators*, `ConZFC_Centered_Residues_2026-10-25.zip`, 25 October 2026. The odd-square radix and compatible parity construction are predecessors to the changed bootstrap in this report.

[4] ProveIt incoming research, *Residual bases and denominator engineering*, `ConZFC_Residual_Bases_Denominator_Engineering_2026-10-25.zip`, 25 October 2026. Source of the exact affine residual exposures used here.

[5] ProveIt direct-calculus source, *Commutative refutation certificates*; retained `COMMUTATIVE_CERTIFICATES.md`, numerical-calculus proofs and calibration interfaces in the source closure of [2].

[6] B. Farhi, *An identity involving the least common multiple of binomial coefficients and its application*, 2009, arXiv:0906.2295v2.

[7] J. Bayer, M. David, M. Hassler, Y. Matiyasevich, and D. Schleicher, *Diophantine equations over $\mathbb{Z}$: universal bounds and parallel formalization*, arXiv:2506.20909v3, 22 October 2026. Background on the domain/degree/variable tradeoff and the distinction between ordinary and formal verification.

A Complete quadratic residuals

All coordinates below are unrestricted integers. An empty monomial in the JSON definition denotes the constant one; repeated coordinate indices denote repeated factors. The list below is generated from the final canonical JSON, and the release verifier checks their agreement. The single equation is exactly the sum of the squares of these residuals.

$$
\begin{aligned}
R_{0} &= -x_{3}^{2} + x_{4} \\
R_{1} &= x_{0} - x_{0}x_{2} - x_{2} + x_{8} \\
R_{2} &= -x_{2}x_{6} - x_{3} + x_{7}x_{8} \\
R_{3} &= -1 + x_{1} - x_{7}x_{8} \\
R_{4} &= 1 + x_{4} - x_{5}x_{19} + x_{17} \\
R_{5} &= -1 - x_{9}^{2} - 2\,x_{9}x_{10} - x_{10}^{2} + x_{20} \\
R_{6} &= -10\,x_{9} + x_{9}^{2} - 6\,x_{9}x_{10} - 8\,x_{9}x_{20} - 20\,x_{10} - 11\,x_{10}^{2} - 16\,x_{10}x_{20} \\
&\quad - 15\,x_{20} - 4\,x_{20}^{2} + x_{21} \\
R_{7} &= -2 + 2\,x_{9} - x_{9}^{2} - 4\,x_{9}x_{21} - 4\,x_{21} - 4\,x_{21}^{2} + x_{22} \\
R_{8} &= 1 + 6\,x_{9} - 36\,x_{9}^{2} - 24\,x_{9}x_{22} + 2\,x_{22} - 4\,x_{22}^{2} + x_{23} \\
R_{9} &= -2 - x_{10} - x_{10}^{2} - 2\,x_{10}x_{11} - x_{11} - x_{11}^{2} + x_{24} \\
R_{10} &= -8 + 7\,x_{9} - x_{9}^{2} + 4\,x_{9}x_{11} - 4\,x_{9}x_{24} - 4\,x_{11} + 4\,x_{24} + x_{25} - x_{47} \\
R_{11} &= -1 + x_{9} - x_{9}^{2} - 2\,x_{9}x_{11} + x_{11} - x_{11}^{2} + x_{26} \\
R_{12} &= 23 + 16\,x_{9} - 8\,x_{9}x_{10} - 8\,x_{9}x_{11} - 16\,x_{9}x_{20} - 8\,x_{9}x_{26} + 19\,x_{10} \\
&\quad + 9\,x_{10}^{2} - 36\,x_{10}x_{11} - 12\,x_{10}x_{26} + 16\,x_{11} + 8\,x_{11}^{2} - 24\,x_{11}x_{20} \\
&\quad - 8\,x_{20}x_{26} - x_{21} + x_{27} - x_{32} \\
R_{13} &= -4\,x_{25}^{2} - 8\,x_{25}x_{27} - 4\,x_{27}^{2} + x_{28} \\
R_{14} &= -2 - 8\,x_{9} + 8\,x_{9}x_{24} + 8\,x_{9}x_{26} - 8\,x_{10} + 8\,x_{10}x_{24} + 8\,x_{10}x_{26} \\
&\quad - 4\,x_{20} + 8\,x_{24} - 4\,x_{24}^{2} - 8\,x_{24}x_{26} + 8\,x_{26} - 4\,x_{26}^{2} + x_{29} \\
R_{15} &= 6 + 4\,x_{9} + 32\,x_{9}x_{10} + 16\,x_{9}x_{20} + 16\,x_{9}x_{26} + 8\,x_{9}x_{29} + 24\,x_{10} \\
&\quad - 32\,x_{10}x_{26} - 16\,x_{10}x_{29} + 12\,x_{20} - 16\,x_{20}x_{26} - 8\,x_{20}x_{29} - x_{21} \\
&\quad + 8\,x_{26} - 16\,x_{26}^{2} - 16\,x_{26}x_{29} + 4\,x_{29} - 4\,x_{29}^{2} + x_{30}
\end{aligned}
$$

$$
\begin{aligned}
R_{16} &= 22 - 5\,x_{9} - x_{9}^{2} - 8\,x_{9}x_{10} - 12\,x_{9}x_{11} - 4\,x_{9}x_{24} - 8\,x_{10} \\
&\quad - 16\,x_{10}x_{11} - 16\,x_{10}x_{24} - 20\,x_{11} - 20\,x_{11}^{2} - 24\,x_{11}x_{24} - 28\,x_{24} \\
&\quad - 4\,x_{24}^{2} + x_{31} \\
R_{17} &= 19 - 25\,x_{9} - 7\,x_{9}^{2} - 8\,x_{9}x_{10} - 30\,x_{9}x_{11} - 16\,x_{9}x_{26} + 3\,x_{10} \\
&\quad + 5\,x_{10}^{2} - 4\,x_{10}x_{26} - 27\,x_{11} - 21\,x_{11}^{2} - 24\,x_{11}x_{26} - 6\,x_{24} - 17\,x_{26} \\
&\quad - 4\,x_{26}^{2} + x_{32} \\
R_{18} &= 2 - 4\,x_{31}^{2} - 8\,x_{31}x_{32} - 4\,x_{32}^{2} + x_{33} \\
R_{19} &= -19\,x_{9} + 8\,x_{9}^{2} + 8\,x_{9}x_{20} + 8\,x_{9}x_{24} - 46\,x_{10} - 24\,x_{10}^{2} \\
&\quad + 24\,x_{10}x_{20} + 24\,x_{10}x_{24} - 2\,x_{11} + 4\,x_{20} - 4\,x_{20}^{2} - 8\,x_{20}x_{24} \\
&\quad + 14\,x_{24} - 4\,x_{24}^{2} - x_{26} + x_{34} \\
R_{20} &= -3 - 6\,x_{9} - 4\,x_{11} - 2\,x_{26} - 4\,x_{34} - 4\,x_{34}^{2} + x_{35} \\
R_{21} &= 8 + 42\,x_{9} + 7\,x_{9}^{2} + 28\,x_{9}x_{21} + 48\,x_{10} - 32\,x_{10}^{2} - 32\,x_{10}x_{20} \\
&\quad + 64\,x_{10}x_{21} - 40\,x_{20}^{2} + 32\,x_{20}x_{21} + 14\,x_{21} - 4\,x_{21}^{2} + x_{36} + 2\,x_{39} \\
R_{22} &= 1 + 6\,x_{9} - 36\,x_{9}^{2} - 24\,x_{9}x_{36} + 2\,x_{36} - 4\,x_{36}^{2} + x_{37} \\
R_{23} &= -x_{10}^{2} - 4\,x_{10}x_{37} - 4\,x_{37}^{2} + x_{38} \\
R_{24} &= -6 - 2\,x_{9} - 4\,x_{10} - 14\,x_{20} - 16\,x_{20}^{2} + x_{39} \\
R_{25} &= 24 + 21\,x_{10} - 21\,x_{10}^{2} - 18\,x_{10}x_{11} - 24\,x_{10}x_{24} - 24\,x_{10}x_{39} + 3\,x_{11} \\
&\quad - 9\,x_{11}^{2} - 16\,x_{11}x_{39} + 3\,x_{24} - 12\,x_{24}^{2} - 8\,x_{24}x_{39} + 4\,x_{39} \\
&\quad - 4\,x_{39}^{2} + x_{40} + 2\,x_{47} \\
R_{26} &= 16 - 10\,x_{9} - 20\,x_{9}^{2} - 16\,x_{9}x_{11} - 24\,x_{9}x_{26} - 24\,x_{9}x_{40} - 12\,x_{11} \\
&\quad - 16\,x_{11}x_{26} - 16\,x_{11}x_{40} - 14\,x_{26} - 4\,x_{26}^{2} - 8\,x_{26}x_{40} + 4\,x_{40} \\
&\quad - 4\,x_{40}^{2} + x_{41} \\
R_{27} &= -x_{11}^{2} - 2\,x_{11}x_{12} - x_{12}^{2} + x_{42} \\
R_{28} &= 2 + x_{9}^{2} - 4\,x_{9}x_{42} + 2\,x_{11} - 2\,x_{11}^{2} - 8\,x_{11}x_{42} - 2\,x_{26} + 4\,x_{42} \\
&\quad - 4\,x_{42}^{2} + x_{43} \\
R_{29} &= -31 - 16\,x_{11} - 16\,x_{11}x_{24} + 16\,x_{11}x_{26} - 8\,x_{24} + 12\,x_{24}^{2} - 8\,x_{24}x_{26} \\
&\quad + 24\,x_{26} - 4\,x_{26}^{2} + x_{44} - 4\,x_{47} \\
R_{30} &= -4\,x_{11} - 16\,x_{26}^{2} - 32\,x_{26}x_{42} - 16\,x_{26}x_{43} - 16\,x_{26}x_{44} - 4\,x_{42} \\
&\quad - 16\,x_{42}^{2} - 16\,x_{42}x_{43} - 16\,x_{42}x_{44} - 2\,x_{43} - 4\,x_{43}^{2} - 8\,x_{43}x_{44} \\
&\quad - 4\,x_{44}^{2} + x_{45} \\
R_{31} &= -16\,x_{9}^{2} + 8\,x_{9}x_{11} + 32\,x_{9}x_{26} + 16\,x_{9}x_{29} + x_{11} - x_{11}^{2} \\
&\quad - 8\,x_{11}x_{26} - 4\,x_{11}x_{29} - 16\,x_{26}^{2} - 16\,x_{26}x_{29} - 4\,x_{29}^{2} + x_{46} \\
R_{32} &= -1 - 8\,x_{11} - 4\,x_{11}^{2} + 8\,x_{11}x_{24} + 8\,x_{24} - 4\,x_{24}^{2} + x_{47} \\
R_{33} &= -1 - 9\,x_{11}^{2} + 24\,x_{11}x_{24} + 12\,x_{11}x_{47} - 16\,x_{24}^{2} - 16\,x_{24}x_{47} \\
&\quad - 4\,x_{47}^{2} + x_{48} \\
R_{34} &= 1 + x_{10} - x_{10}^{2} - 4\,x_{10}x_{11} - 4\,x_{10}x_{48} - 4\,x_{11}^{2} - 8\,x_{11}x_{48} \\
&\quad - 4\,x_{48}^{2} + x_{49} \\
R_{35} &= -1 - 4\,x_{49} - 4\,x_{49}^{2} + x_{50} \\
R_{36} &= -1 - x_{11}^{2} - 8\,x_{11}x_{49} - 4\,x_{11}x_{50} - 16\,x_{49}^{2} - 16\,x_{49}x_{50} - 4\,x_{50}^{2} \\
&\quad + x_{51} \\
R_{37} &= 2 + x_{10} - x_{10}^{2} - 4\,x_{10}x_{11} - 4\,x_{10}x_{51} - 4\,x_{11}^{2} - 8\,x_{11}x_{51} \\
&\quad - 4\,x_{51}^{2} + x_{52} \\
R_{38} &= -4\,x_{46}^{2} - 8\,x_{46}x_{52} - 4\,x_{52}^{2} + x_{53} \\
R_{39} &= -9 + 7\,x_{9} - x_{9}^{2} - 8\,x_{9}x_{46} - 4\,x_{9}x_{53} + 24\,x_{46} - 16\,x_{46}^{2} \\
&\quad - 16\,x_{46}x_{53} + 12\,x_{53} - 4\,x_{53}^{2} + x_{54} \\
R_{40} &= -1 - 4\,x_{54} - 4\,x_{54}^{2} + x_{55}
\end{aligned}
$$

$$
\begin{aligned}
R_{41} =& -x_{11}^2 - 8\,x_{11}x_{54} - 4\,x_{11}x_{55} - 16\,x_{54}^2 - 16\,x_{54}x_{55} - 4\,x_{55}^2 \\
&+ x_{56} \\
R_{42} =& -x_9 - x_9^2 - 4\,x_9x_{11} - 4\,x_9x_{56} - 4\,x_{11} - 4\,x_{11}^2 - 8\,x_{11}x_{56} \\
&- 4\,x_{56} - 4\,x_{56}^2 + x_{57} \\
R_{43} =& -10 - 12\,x_{57} - 4\,x_{57}^2 + x_{58} \\
R_{44} =& \,2 + 4\,x_{11} + 16\,x_{11}x_{57} + 8\,x_{11}x_{58} - 4\,x_{24} - 16\,x_{24}x_{57} - 8\,x_{24}x_{58} \\
&- x_{47} + 8\,x_{57} - 16\,x_{57}^2 - 16\,x_{57}x_{58} + 4\,x_{58} - 4\,x_{58}^2 + x_{59} \\
R_{45} =& \,x_{10} - x_{10}^2 + 8\,x_{10}x_{11} - 8\,x_{10}x_{24} - 4\,x_{10}x_{59} - 16\,x_{11}^2 \\
&+ 32\,x_{11}x_{24} + 16\,x_{11}x_{59} - 16\,x_{24}^2 - 16\,x_{24}x_{59} - 4\,x_{59}^2 + x_{60} \\
R_{46} =& -10 - 12\,x_{52} - 4\,x_{52}^2 + x_{61} \\
R_{47} =& \,1 - 16\,x_{52}^2 - 16\,x_{52}x_{60} - 16\,x_{52}x_{61} - 4\,x_{60}^2 - 8\,x_{60}x_{61} - 4\,x_{61}^2 \\
&+ x_{62} \\
R_{48} =& \,x_{11} - x_{11}^2 - 8\,x_{11}x_{60} - 4\,x_{11}x_{62} - 16\,x_{60}^2 - 16\,x_{60}x_{62} \\
&- 4\,x_{62}^2 + x_{63} \\
R_{49} =& -6\,x_9 - 3\,x_9^2 - 18\,x_9x_{10} - 16\,x_{10} - 27\,x_{10}^2 + 4\,x_{10}x_{11} + 3\,x_{11} \\
&+ 3\,x_{11}^2 - 4\,x_{11}x_{20} - 19\,x_{20} - 8\,x_{20}^2 + x_{21} + x_{64} - 4\,x_{77} \\
R_{50} =& \,1 - 8\,x_9 - 4\,x_9x_{11} + 8\,x_9x_{64} - 16\,x_{10} - 8\,x_{10}x_{11} + 16\,x_{10}x_{64} + x_{11} \\
&+ x_{11}^2 + 4\,x_{11}x_{20} + 8\,x_{20} - 8\,x_{20}x_{64} + 3\,x_{64} - 4\,x_{64}^2 + x_{65} \\
R_{51} =& -2 - 12\,x_9 - 16\,x_9x_{11} + 16\,x_9x_{63} + 16\,x_9x_{65} - 24\,x_{10} - 32\,x_{10}x_{11} \\
&+ 32\,x_{10}x_{63} + 32\,x_{10}x_{65} + 4\,x_{11} + 4\,x_{11}^2 + 16\,x_{11}x_{20} + 12\,x_{20} \\
&- 16\,x_{20}x_{63} - 16\,x_{20}x_{65} - 4\,x_{63}^2 - 8\,x_{63}x_{65} - 4\,x_{64} - 2\,x_{65} \\
&- 4\,x_{65}^2 + x_{66} \\
R_{52} =& -2 - 4\,x_{66} - 4\,x_{66}^2 + x_{67} \\
R_{53} =& -x_3x_{14} + x_{68} \\
R_{54} =& \,x_{13} - x_{14} - 2\,x_{23}x_{68} - x_{69} \\
R_{55} =& \,x_{13} - x_{14} - 4\,x_{25}x_{68} - 2\,x_{28}x_{68} - x_{70} \\
R_{56} =& -2\,x_{30}x_{68} + 2\,x_{45}x_{68} - 2\,x_{68} - x_{71} + x_{76} \\
R_{57} =& \,x_{13} - x_{14} - 4\,x_{31}x_{68} - 2\,x_{33}x_{68} - x_{72} \\
R_{58} =& \,x_{13} - x_{14} - 2\,x_{35}x_{68} + 6\,x_{68} - x_{73} \\
R_{59} =& \,x_{13} - x_{14} - 4\,x_{37}x_{68} - 2\,x_{38}x_{68} - x_{74} \\
R_{60} =& \,x_{13} - x_{14} - 2\,x_{41}x_{68} - x_{75} \\
R_{61} =& \,x_{13} - x_{14} - 2\,x_{45}x_{68} - 4\,x_{68} - x_{76} \\
R_{62} =& -x_9x_{10} + x_9x_{11} + x_{10} + x_{10}x_{11} - x_{11} - x_{11}^2 + x_{77} \\
R_{63} =& \,1 + 2\,x_{12} + x_{12}^2 - x_{77}^2 + x_{78} \\
R_{64} =& -x_{78}^2 + x_{79} \\
R_{65} =& -4\,x_{66}x_{68} - 2\,x_{67}x_{68} + 2\,x_{68} + x_{80} \\
R_{66} =& -x_{13}^2 + 2\,x_{13}x_{14} + 2\,x_{13}x_{80} - x_{14}^2 - 2\,x_{14}x_{80} - x_{80}^2 + x_{81} \\
R_{67} =& -1 - x_3x_{10} + x_{82} \\
R_{68} =& -x_{14}x_{82} + x_{84}x_{85} + x_{86} \\
R_{69} =& -1 + x_3 - x_3x_9 + x_{83} \\
R_{70} =& -1 + 2\,x_3 + 2\,x_3x_9 + 4\,x_3x_{10} - 2\,x_3x_{20} + x_{84} \\
R_{71} =& -x_{13}x_{83} + x_{85} \\
R_{72} =& -6\,x_3x_{14} + x_{13}x_{82} + 12\,x_{14} - 8\,x_{14}x_{82} - 4\,x_{14}x_{83} - x_{14}x_{84} + x_{87} \\
R_{73} =& -x_{13}x_{69} + x_{14}x_{69} + x_{88} \\
R_{74} =& -x_{70}x_{88} + x_{89} \\
R_{75} =& -x_{71}x_{89} + x_{90} \\
R_{76} =& -x_{72}x_{90} + x_{91}
\end{aligned}
$$

$$
\begin{aligned}
R_{77} &= -x_{73}x_{91} + x_{92} \\
R_{78} &= -x_{74}x_{92} + x_{93} \\
R_{79} &= -x_{75}x_{93} + x_{94} \\
R_{80} &= -x_{76}x_{94} + x_{95} \\
R_{81} &= -x_{86}x_{95} + x_{96} \\
R_{82} &= -x_{87}x_{96} + x_{97} \\
R_{83} &= -x_{97}^2 + x_{98} \\
R_{84} &= -x_2 - x_2x_{17} + x_{17} + x_{99} \\
R_{85} &= -x_{15}x_{99} + x_{100} \\
R_{86} &= -2\,x_1^2 + 2\,x_1x_{13} + 2\,x_1x_{14} + 2\,x_1x_{100} + 2\,x_1x_{101} - x_{13}^2 \\
&\quad - 2\,x_{13}x_{100} - x_{14}^2 - 2\,x_{14}x_{101} + x_{18}x_{103} - x_{79}x_{98} - x_{81}x_{98} \\
&\quad - x_{100}^2 - x_{101}^2 \\
R_{87} &= -x_2x_{16} + x_{16} - x_{16}x_{99} + x_{101} \\
R_{88} &= -x_9 - x_{10} - 4\,x_{20} - 2\,x_{23} - 4\,x_{25} - 2\,x_{28} - 2\,x_{30} - 4\,x_{31} - 2\,x_{33} - 2\,x_{35} \\
&\quad - 4\,x_{37} - 2\,x_{38} - 2\,x_{41} - 2\,x_{45} - 4\,x_{66} - 2\,x_{67} + x_{102} \\
R_{89} &= -x_{104}x_{107} + x_{107} + x_{109} \\
R_{90} &= -x_5x_{131} + x_{105}x_{230} - x_{108} - x_{109}x_{131} \\
R_{91} &= x_{110}x_{231} - x_{112}x_{134} - x_{113} \\
R_{92} &= x_{114}x_{231} - x_{115}x_{134} - x_{116} \\
R_{93} &= x_{117}x_{231} - x_{118}x_{134} - x_{119} \\
R_{94} &= x_{120}x_{231} - x_{121}x_{134} - x_{122} \\
R_{95} &= x_{123}x_{232} - x_{125}x_{134} - x_{126} \\
R_{96} &= x_{127}x_{232} - x_{128}x_{134} - x_{129} \\
R_{97} &= -x_{18}x_{131} + x_{134} \\
R_{98} &= x_{130} - x_{132}x_{134} - x_{133} \\
R_{99} &= -x_{134}x_{136} + x_{135} - x_{137} \\
R_{100} &= -x_{134}x_{139} + x_{138}x_{230} - x_{140} \\
R_{101} &= -x_{134}x_{142} + x_{141}x_{231} - x_{143} \\
R_{102} &= x_{144} - 2\,x_{144}x_{146} + x_{147} \\
R_{103} &= -x_{144}x_{147} - x_{147} + x_{148} \\
R_{104} &= x_{147}^2 - x_{148}^2 + x_{168} \\
R_{105} &= -x_{145}x_{150} - x_{150} + x_{151} \\
R_{106} &= 2\,x_{149} + x_{149}^2 + x_{150}^2 - x_{151}^2 \\
R_{107} &= 1 + 4\,x_0 + 4\,x_0^2 - 4\,x_0x_{145} - 4\,x_{145} + x_{152} \\
R_{108} &= -1 - 2\,x_0x_{150} + x_3 + x_{145}x_{150} - x_{149} - x_{150} + x_{152}x_{153} \\
R_{109} &= 1 - 2\,x_5 + x_5^2 - 2\,x_5x_{145} + x_{154} \\
R_{110} &= -x_5x_{199} + x_{18} + x_{154}x_{155} - x_{198} + x_{200} \\
R_{111} &= -1 + 2\,x_{145}x_{214} - x_{156} + 2\,x_{214} - x_{214}^2 \\
R_{112} &= -1 + x_{104} + x_{156}x_{157} - x_{198} - x_{199}x_{214} + x_{200} \\
R_{113} &= -x_{145}x_{159} - x_{159} + x_{160} \\
R_{114} &= 2\,x_{158} + x_{158}^2 + x_{159}^2 - x_{160}^2 \\
R_{115} &= 1 - 2\,x_{104} + x_{104}^2 - 2\,x_{104}x_{145} + x_{161} \\
R_{116} &= -1 - x_{104}x_{159} + x_{105} - x_{158} + x_{160} + x_{161}x_{162} \\
R_{117} &= -1 - x_{104}x_{150} + 2\,x_{104}x_{163} + x_{106} - 2\,x_{145}x_{163} - x_{149} + x_{150} + x_{151} \\
&\quad + x_{161}x_{163} - 3\,x_{163} \\
R_{118} &= 1 + x_{145} - x_{145}x_{165} - x_{165} + x_{166} \\
R_{119} &= 1 + 2\,x_{164} + x_{164}^2 - 2\,x_{165} + x_{165}^2 - x_{166}^2
\end{aligned}
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\begin{aligned}
R_{120} &= -1 + x_{104} - x_{104}x_{165} + x_{110} + x_{161}x_{167} - x_{164} + x_{166} \\
R_{121} &= -2\,x_{145} - x_{145}^2 + x_{168} \\
R_{122} &= 1 - x_{104} - x_{104}x_{149} - x_{104}x_{151} + 2\,x_{104}x_{169} + x_{111} + x_{145}x_{151} \\
&\quad - 2\,x_{145}x_{169} + x_{149} - x_{150}x_{168} + 2\,x_{151} + x_{161}x_{169} - 3\,x_{169} \\
R_{123} &= -x_{145}x_{171} - x_{171} + x_{172} \\
R_{124} &= 2\,x_{170} + x_{170}^2 + x_{171}^2 - x_{172}^2 \\
R_{125} &= -1 - x_{104}x_{171} + x_{114} + x_{161}x_{173} - x_{170} + x_{172} \\
R_{126} &= -x_{145}x_{175} - x_{175} + x_{176} \\
R_{127} &= 2\,x_{174} + x_{174}^2 + x_{175}^2 - x_{176}^2 \\
R_{128} &= -1 - x_{104}x_{175} + x_{117} + x_{161}x_{177} - x_{174} + x_{176} \\
R_{129} &= -x_{145}x_{179} - x_{179} + x_{180} \\
R_{130} &= 2\,x_{178} + x_{178}^2 + x_{179}^2 - x_{180}^2 \\
R_{131} &= -1 - x_{104}x_{179} + x_{120} + x_{161}x_{181} - x_{178} + x_{180} \\
R_{132} &= -x_{145}x_{183} - x_{183} + x_{184} \\
R_{133} &= 2\,x_{182} + x_{182}^2 + x_{183}^2 - x_{184}^2 \\
R_{134} &= -1 - x_{104}x_{183} + x_{123} + x_{161}x_{185} - x_{182} + x_{184} \\
R_{135} &= -x_{145}x_{187} - x_{187} + x_{188} \\
R_{136} &= 2\,x_{186} + x_{186}^2 + x_{187}^2 - x_{188}^2 \\
R_{137} &= -1 - x_{104}x_{187} + 2\,x_{104}x_{189} + x_{124} - 2\,x_{145}x_{189} + x_{161}x_{189} - x_{186} + x_{187} \\
&\quad + x_{188} - 3\,x_{189} \\
R_{138} &= -x_{145}x_{191} - x_{191} + x_{192} \\
R_{139} &= 2\,x_{190} + x_{190}^2 + x_{191}^2 - x_{192}^2 \\
R_{140} &= -1 - x_{104}x_{191} + x_{127} + x_{161}x_{193} - x_{190} + x_{192} \\
R_{141} &= -x_{145}x_{195} - x_{195} + x_{196} \\
R_{142} &= 2\,x_{194} + x_{194}^2 + x_{195}^2 - x_{196}^2 \\
R_{143} &= -1 - x_{104}x_{195} + x_{130} + x_{161}x_{197} - x_{194} + x_{196} \\
R_{144} &= 1 - x_{145}x_{199} - x_{199} + x_{200} \\
R_{145} &= -1 + 2\,x_{198} + x_{198}^2 + x_{199}^2 - 2\,x_{200} - x_{200}^2 \\
R_{146} &= -x_{104}x_{199} + 2\,x_{104}x_{201} + x_{131} - 2\,x_{145}x_{201} + x_{161}x_{201} - x_{198} + x_{199} \\
&\quad + x_{200} - 3\,x_{201} \\
R_{147} &= -x_{145}x_{203} - x_{203} + x_{204} \\
R_{148} &= 2\,x_{202} + x_{202}^2 + x_{203}^2 - x_{204}^2 \\
R_{149} &= -1 - x_{104}x_{203} + x_{135} + x_{161}x_{205} - x_{202} + x_{204} \\
R_{150} &= -x_{145}x_{207} - x_{207} + x_{208} \\
R_{151} &= 2\,x_{206} + x_{206}^2 + x_{207}^2 - x_{208}^2 \\
R_{152} &= -1 - x_{104}x_{207} + x_{138} + x_{161}x_{209} - x_{206} + x_{208} \\
R_{153} &= -x_{145}x_{211} - x_{211} + x_{212} \\
R_{154} &= 2\,x_{210} + x_{210}^2 + x_{211}^2 - x_{212}^2 \\
R_{155} &= -1 - x_{104}x_{211} + x_{141} + x_{161}x_{213} - x_{210} + x_{212} \\
R_{156} &= -8 - 10\,x_0 - 7\,x_0^2 - 2\,x_0x_4 - x_1^2 - x_2^2 - x_3^2 - x_4^2 - x_5^2 \\
&\quad + 2\,x_9 + 2\,x_9x_{10} - x_{11}^2 - x_{12}^2 - x_{13}^2 - x_{14}^2 - x_{15}^2 - x_{16}^2 \\
&\quad - x_{17}^2 - x_{18}^2 - x_{20} - x_{102}^2 + 2\,x_{104} - 2\,x_{104}^2 - x_{105}^2 - x_{106}^2 \\
&\quad - x_{110}^2 - x_{111}^2 - x_{114}^2 - x_{117}^2 - x_{120}^2 - x_{123}^2 - x_{124}^2 - x_{127}^2 \\
&\quad - x_{130}^2 - x_{131}^2 - x_{135}^2 - x_{138}^2 - x_{141}^2 + x_{144} - x_{214}^2 \\
R_{157} &= -x_5 - x_5x_9 - x_5x_{10} - x_5x_{11} - x_5x_{12} - x_5x_{13} - x_5x_{14} - x_5x_{15} \\
&\quad - x_5x_{16} - x_5x_{17} + x_{214} \\
R_{158} &= -16 + x_0 - x_{215}^2
\end{aligned}
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\begin{aligned}
R_{159} =& -x_{145}x_{217} - x_{217} + x_{226} \\
R_{160} =& -1 + x_{216}^2 + x_{217}^2 - x_{226}^2 \\
R_{161} =& -x_{219} - x_{219}x_{220} + x_{227} \\
R_{162} =& -1 + x_{218}^2 + x_{219}^2 - x_{227}^2 \\
R_{163} =& -x_{211}^2 + x_{228} \\
R_{164} =& \, x_{217} - 2\,x_{222}x_{228} - x_{228} \\
R_{165} =& -4\,x_{217} - 8\,x_{217}x_{221} + x_{220} \\
R_{166} =& -x_{145} - x_{216}x_{223} + x_{220} \\
R_{167} =& -1 - x_{210} - x_{216}x_{224} + x_{218} \\
R_{168} =& -x_{102} - 4\,x_{211}x_{225} + x_{219} \\
R_{169} =& -2 + 2\,x_0^2 + 2\,x_0x_4 - 2\,x_0x_{150} - 2\,x_0x_{159} + x_1^2 - 2\,x_1x_{199} \\
& + x_2^2 - 2\,x_2x_{187} + x_4^2 - 2\,x_4x_{159} + x_9^2 - 2\,x_9x_{165} + x_{10}^2 \\
& - 2\,x_{10}x_{171} + x_{11}^2 - 2\,x_{11}x_{175} + x_{12}^2 - 2\,x_{12}x_{179} + x_{13}^2 \\
& - 2\,x_{13}x_{183} + x_{14}^2 - 2\,x_{14}x_{191} + x_{15}^2 - 2\,x_{15}x_{195} + x_{16}^2 \\
& - 2\,x_{16}x_{203} + x_{17}^2 - 2\,x_{17}x_{207} - 2\,x_{102}x_{211} - x_{145}x_{229} + x_{149}^2 \\
& + x_{158}^2 + x_{164}^2 + 2\,x_{165} + x_{170}^2 + x_{174}^2 + x_{178}^2 + x_{182}^2 + x_{186}^2 \\
& + x_{190}^2 + x_{194}^2 + x_{198}^2 + x_{202}^2 + x_{206}^2 + x_{210}^2 + 2\,x_{211}^2 \\
& - x_{233}^2 - x_{234}^2 - x_{235} - x_{235}^2 \\
R_{170} =& -x_{106}x_{230} + x_{131} \\
R_{171} =& -x_{111}x_{231} + x_{131} \\
R_{172} =& -x_{124}x_{232} + x_{131} \\
R_{173} =& -1 + x_2 - x_3x_{240} \\
R_{174} =& -2 - x_{102}^2 - x_{150}^2 - x_{159}^2 + 2\,x_{165} - x_{165}^2 - x_{171}^2 - x_{175}^2 \\
& - x_{179}^2 - x_{183}^2 - x_{187}^2 - x_{191}^2 - x_{195}^2 - x_{199}^2 - x_{203}^2 - x_{207}^2 \\
& + x_{211}^2 - x_{233}^2 - x_{234}^2 - x_{235} - x_{235}^2 \\
R_{175} =& -1 - 4\,x_{108}^2 - 4\,x_{113}^2 - 4\,x_{116}^2 - 4\,x_{119}^2 - 4\,x_{122}^2 - 4\,x_{126}^2 \\
& - 4\,x_{129}^2 + x_{131}^2 - 4\,x_{133}^2 - 4\,x_{137}^2 - 4\,x_{140}^2 - 4\,x_{143}^2 \\
& - x_{236}^2 - x_{237}^2 - x_{238} - x_{238}^2 \\
R_{176} =& -2 + x_{104} - 4\,x_{239} - 4\,x_{239}^2 \\
R_{177} =& -1 + x_{240} - 4\,x_{241} - 4\,x_{241}^2
\end{aligned}
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